FINAL YEAR MAJOR PROJECT PRESENTATION

CampusPulse

Enterprise Campus Event Discovery, Real-Time Engagement & Institutional Analytics Platform

A Next.js 16 & MongoDB Atlas production ecosystem featuring simple 1-click registration, live Q&A with upvoting, dynamic feedback surveys, and automated executive PDF accreditation reports.

SUBMITTED BY (STUDENT TEAM):

[STUDENT 1 NAME] — Roll No: [ROLL NUMBER]
[STUDENT 2 NAME] — Roll No: [ROLL NUMBER]
Department of Computer Science & Engineering
[INSTITUTE / UNIVERSITY NAME], Year 2025-2026

PROJECT GUIDANCE & MENTORSHIP:

Under the Supervision of:
[MENTOR / PROFESSOR NAME]
[DESIGNATION, e.g. Assistant Professor / HOD]
Department of Computer Science & Engineering

Problem Statement & Campus Challenges

1. Scattered & Fragmented Notice Distribution:
Event notices get buried across scattered WhatsApp groups, cluttered inboxes, and physical notice boards. Students miss registration deadlines and venue announcements.

2. Inefficient & Manual Registration Queues:
Paper forms and manual spreadsheet entries create massive desk bottlenecks, handwriting errors, lost registrations, and frequent seat overbooking.

3. Low Student Feedback Participation:
Traditional post-event Google Forms see less than 10% student participation with zero automated sentiment analysis, leaving organizers blind to attendee satisfaction.

4. Heavy Accreditation (NAAC/NBA) Documentation:
Faculty coordinators spend weeks compiling attendance records, photos, and outcomes manually for institutional accreditation audits.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT / GRAPHIC]

Diagram showing chaotic campus paper notices, fragmented WhatsApp circulars, and long manual registration queues.

Proposed Solution & Project Objectives

1. Centralized Campus Event Hub:
A single unified responsive web and mobile portal aggregating Technical, Cultural, Sports, and Academic Workshop events across all departments.

2. Simple 1-Click RSVP with Capacity Control:
Instant seat reservation backed by atomic database locks to prevent overbooking, accompanied by an instant digital Registration Confirmation Card.

3. Interactive In-Session Student Q&A:
Live session engagement board with atomic deduplicated student upvotes, allowing trending questions to rise to the speaker's attention.

4. Dynamic Feedback & Automated Accreditation Reports:
Google-Forms-style survey builder with real-time Net Promoter Score (NPS) analytics and 1-Click Executive PDF Letterhead reports for college administration.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT / DIAGRAM]

CampusPulse high-level ecosystem diagram connecting Students, Event Coordinators, Live Q&A, Feedback, and Admin Reports.

System Architecture & Technology Stack

- **Frontend Layer:**
  Next.js 16.3.3 (Turbopack, App Router), React 19.2.8, Tailwind CSS v4, Framer Motion for smooth micro-animations, Lucide React icons.

- **Backend & Server Actions:**
  Next.js Server Actions (src/actions/) with 50MB body streaming, Edge Route Handlers, role-based middleware guards.

- **Database & ORM:**
  MongoDB Atlas Cloud, Mongoose 9.9.4 with Global Connection Caching Singleton and Google Public DNS (8.8.8.8) SRV resolver.

- **Authentication & Security:**
  NextAuth.js v4 (Stateless JWT sessions, Bcrypt password hashing with 10 salt rounds, strict student/admin RBAC).

- **Mobile & Deployment:**
  Capacitor 8.5 (Native Android APK), Netlify Serverless with automated GitHub webhooks, Multi-stage Docker container.

IMAGE / SCREENSHOT

[INSERT ARCHITECTURE DIAGRAM]

Three-tier architecture diagram showing Client (Web & Android), Next.js 16 Server Actions / NextAuth, and MongoDB Atlas Persistence layer.

Module 1: Event Discovery & Categorized Directory

- **Multi-Category Academic Catalog:**
  Events categorized into Technical (Hackathons, Coding), Cultural (Fests, Music), Sports (Tournaments), and Academic Workshops.

- **Real-Time Fuzzy Search & Filtering:**
  Students can instantly search by title, department, tags, or venue with dynamic status toggles (Upcoming, Ongoing, Completed).

- **Rich Event Detail View:**
  Each event showcases high-res banner media, detailed schedules, venue location, faculty coordinator contacts, and prerequisites.

- **1-Click Viral Social Sharing:**
  Deep-linked sharing modal for WhatsApp, Telegram, LinkedIn, and X/Twitter with pre-formatted event titles and registration links.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: /events]

Screenshot of the CampusPulse Event Discovery Catalog showing category filter pills, search bar, and colorful event cards.

Module 2: Simple 1-Click Registration & Capacity Control

- **Frictionless 1-Click Registration:**
  Logged-in students click 'Register for Event' and are instantly enrolled without filling repetitive personal details or manual forms.

- **Atomic Capacity Lock (Zero Overbooking):**
  Database transactions atomically enforce seat limits (maxCapacity). As soon as capacity is filled, the button locks to 'Event Full'.

- **Real-Time Headcount Progress Bar:**
  Visual progress bar showing live filled seats (e.g. '142 / 150 Seats Reserved') with dynamic color shifts as seats run low.

- **Registration Details Card & Cancellation:**
  Confirmed attendees view their name, email, seat status, and unique reference ID. Flexible 1-click cancellation immediately frees up seats.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: /events/[slug]]

Screenshot of the Event Detail Page showing the Registration Confirmed Card, Student Details, and the Capacity Progress Bar.

Module 3: Live Interactive Q&A & Upvoting Forum

- **In-Session Student Queries:**
  During live seminars, hackathons, and guest lectures, attendees post questions from their laptops or phones without interrupting the speaker.

- **Atomic Deduplicated Upvoting:**
  Students upvote questions they care about. MongoDB $addToSet and $pull operations strictly ensure one upvote per student per question.

- **Popularity Ranking Engine:**
  Questions are dynamically sorted by upvote count, ensuring the most urgent student queries naturally rise to the coordinator's screen.

- **Organizer Pinning & Verified Replies:**
  Event coordinators and professors can pin important announcements (isPinned: true) and post official verified answers.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: /portal/qna]

Screenshot of the Live Q&A forum showing student questions, upvote counter buttons, and an organizer's pinned answer badge.

Module 4: Dynamic Feedback & NPS Survey Analytics

- **Google-Forms-Style Dynamic Builder:**
  Coordinators can create custom post-event feedback forms with Star Ratings (1–5), Multiple Choice questions, and Open Text feedback.

- **Automated Net Promoter Score (NPS):**
  The platform automatically aggregates ratings to calculate event NPS (Promoters vs Detractors) and overall average satisfaction scores.

- **Student Anonymity Option:**
  Students can choose to submit verified or anonymous feedback, encouraging honest qualitative critique without fear.

- **Response Analytics & CSV Export:**
  Real-time visualization of participant sentiment breakdowns, attendance-to-response conversion rates, and exportable tables.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: Feedback Analytics]

Screenshot of the student feedback survey form alongside the admin sentiment analytics bar chart and star rating distribution.

Module 5: Media Vault & Executive Accreditation Reports

- **High-Resolution Media Vault:**
  Centralized photo gallery with full-screen lightbox inspection and MP4/WebM video clip playback with YouTube/Vimeo embed parsing.

- **High-Reliability Upload Pipeline:**
  Stores media in local /public/uploads with an automatic Base64 Data URI fallback on read-only serverless clouds so uploads never fail.

- **Automated Executive Report Generator:**
  Automatically compiles Chief Guest details, faculty coordinators, objectives, student attendance numbers, and feedback stats.

- **1-Click Institutional PDF Export:**
  Powered by @react-pdf/renderer, coordinators export clean, official letterhead PDF documents ready for NAAC/NBA audit files.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: Executive PDF Report]

Screenshot of the Media Vault Photo Gallery alongside a sample generated Institutional Executive PDF Report on college letterhead.

Native Android Mobile Application (Capacitor 8.5)

- **Hybrid Native Architecture:**
  Packaged using Capacitor 8.5 into a full-screen, hardware-accelerated Android WebView with native lifecycle controls.

- **Standalone Production Binary (CampusPulse.apk):**
  Pre-compiled APK (Application ID: com.campuspulse.app, Size: ~3.68 MB) ready for direct sideloading and Play Store distribution.

- **Native Assets & Brand Isolation:**
  Custom branded splash screens (splash.png) and adaptive launcher icons across all Android DPI buckets (hdpi to xxxhdpi).

- **Live Sync Cloud Updates:**
  Connects to production remote URL (APP_URL), enabling web updates to reflect instantly on mobile devices without requiring app store updates.

IMAGE / SCREENSHOT

[INSERT SCREENSHOT: Android App Mockup]

Photo of CampusPulse running on an Android smartphone screen, showing the mobile navigation drawer and event cards.

Security, RBAC & Multi-Platform DevOps

- **Role-Based Access Control (RBAC):**
  Server-level guards (requireAdmin(), requireAuth()) strictly protect admin panels, survey creators, and report export endpoints.

- **Cryptographic Session Security:**
  Bcrypt password hashing with 10 salt rounds, stateless 30-day JWT sessions, and security headers (nosniff, X-Frame-Options, XSS protection).

- **100% Free Serverless Hosting (Netlify):**
  Fully configured netlify.toml with automated GitHub webhook CI/CD pipeline. Pushing to origin/main triggers instant live deployment.

- **Production Multi-Stage Dockerization:**
  Lightweight Dockerfile (Node 20 Alpine) with non-root user and docker-compose.yml with persistent volume mounting for VPS hosting.

IMAGE / SCREENSHOT

[INSERT DEVOPS FLOWCHART]

Diagram showing the CI/CD pipeline: Git Commit -> GitHub Push -> Netlify Automatic Webhook Build -> Live Web & Mobile App.

Project Impact, Future Scope & Conclusion

MEASURABLE PROJECT IMPACT

- **100% Paperless Registration:**
  Completely eliminates manual paper forms and physical desk queues.
- **5-Second Seat Confirmation:**
  Atomic database reservation guarantees zero seat overbooking.
- **300% Boost in Student Feedback:**
  In-platform survey forms with automated sentiment scoring and NPS.
- **90% Administrative Time Saved:**
  Generates institutional accreditation audit reports in 1 click.

FUTURE ENHANCEMENTS & ROADMAP

- **AI Event Recommendation Engine:**
  Suggests seminars based on student department & past preferences.
- **Automated Participation Certificates:**
  Digitally signed PDF certificates generated automatically post-event.
- **Campus Push & SMS Notifications:**
  Real-time alerts for schedule delays and venue changes.

THANK YOU!

Questions & Open Discussion with the Panel